

VPA ICT — Grades 6–8 Assessment Pack

Grades 6 – 8 Assessment Pack

Program: Digital Mastery | Assessment Types: Formative, Summative, Portfolio

Assessment Philosophy

Middle school assessment bridges elementary portfolios and IGCSE examinations. Students receive rubric-based feedback for every project, complete timed practical tasks, and build a comprehensive digital portfolio. Coding assessments use auto-grading platforms (Replit, CodeHS) for immediate feedback with teacher verification.

Year 6 Assessment — Advanced Productivity & Web Design

Criterion	Bronze	Silver	Gold
Word Processing	Uses styles for 2 heading levels	Styles + auto-generated TOC + mail merge	Complex multi-section document with different headers/footers
Spreadsheets	Uses 2 functions (SUM, AVERAGE)	IF functions + conditional formatting + data validation	Nested IFs, lookup functions, protected cells, professional dashboard
Web Design	2 pages with basic HTML tags	3 pages with internal CSS and navigation	3+ pages with external CSS, images, tables, forms
Problem Solving	Follows tutorials with support	Adapts examples to new situations	Creates original solutions independently

Year 7 Assessment — Programming & Databases

Criterion	Bronze	Silver	Gold
Python Syntax	Runs basic programs with occasional errors	Consistent syntax, uses comments	Clean, well-commented code with efficient logic
Variables & Data Types	Uses strings and integers	Uses strings, integers, floats, booleans appropriately	Uses lists, dictionaries; type-aware design
Control Structures	Uses if/else or loops separately	Combines if/else with loops	Nested logic, complex conditions, efficient loops
Functions	Uses one simple function	Multiple functions with parameters	Modular design with reusable functions
Databases	Creates table, enters data	Queries with filters and sorting	Multiple tables, relationships, complex queries
Debugging	Fixes errors with teacher help	Independently finds and fixes syntax errors	Anticipates errors, uses testing strategies

Year 8 Assessment — Cybersecurity, Robotics & Media

Criterion	Bronze	Silver	Gold
Cybersecurity Knowledge	Names 3 threats	Explains threats and protections with examples	Analyzes real case studies; proposes defense strategies
Robotics — Hardware	Connects components with guidance	Builds circuit independently from diagram	Modifies circuit design; troubleshoots independently
Robotics — Programming	Uploads example code	Modifies example code for new behavior	Writes original program from pseudocode
Video Editing	3 clips with basic cuts	5+ clips with transitions, titles, audio	Professional quality with effects, color correction, storytelling
Data Visualization	1 basic chart	Multiple chart types with labels and titles	Interactive or automated dashboard; insightful analysis
Collaboration	Participates in group work	Contributes equally, communicates clearly	Leads team, resolves conflicts, mentors peers

Coding Assessment Rubric (Auto-Graded + Teacher)

Auto-Graded (by platform):

- ☐ Code compiles/runs without errors
- ☐ Passes provided test cases
- ☐ Runs within time limits

Teacher-Graded:

- ☐ Variable names are meaningful (not x, y, z)
- ☐ Code includes comments explaining logic
- ☐ Solution is efficient (not overly complex)
- ☐ Handles edge cases (empty input, zero, negatives)

Portfolio Requirements (End of Grade 8)

Students submit a digital portfolio containing:

- ☐ Best word-processed document (with styles and TOC)
- ☐ Best spreadsheet (with formulas, charts, validation)
- ☐ Best presentation (10+ slides, embedded media)
- ☐ Python project with source code and README
- ☐ Website (HTML/CSS files or hosted link)
- ☐ Database file with documentation
- ☐ Robotics project video or photos
- ☐ Cybersecurity guide or infographic
- ☐ Edited video (1+ minute)

☐ Self-reflection essay (200 words): "My ICT Journey"

IGCSE ICT Readiness Check (End of Grade 8)

Skill	Not Ready	Developing	Ready
Can create a formatted document in under 30 min	☐	☐	☐
Can build a spreadsheet with formulas and charts	☐	☐	☐
Can explain what a database is and why it's used	☐	☐	☐
Can write a simple program in Python	☐	☐	☐
Can create a basic webpage with HTML/CSS	☐	☐	☐
Understands online safety and digital ethics	☐	☐	☐

EAL Support in Middle School Assessment

Bilingual technical glossaries: Provided for all units with English/中文/Pinyin

Scaffolded rubrics: Bronze/Silver/Gold descriptions include visual exemplars

Oral defense option: Students may explain code/projects orally if writing is a barrier

Extended time: +15 minutes for practical assessments

Peer translation: Bilingual students may assist with instruction (not assessment)

Code comments in L1: Students may write comments in Mandarin to demonstrate understanding